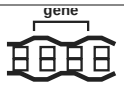
The Siamese Cat Mystery

Focus question: Why are a Siamese cat's ears, paws, and tail dark?

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)	Example	Drawing
DNA	The molecule in every cell that stores the instructions for building proteins.	A recipe book.	The DNA in every cell of the cat.	
gene	A section of DNA with the instructions for one protein.	One recipe in the book.	The gene for the pigment-making protein.	
protein	A molecule built from amino acids that does a job for the cell.	A worker or a machine.	The protein that makes dark pigment.	

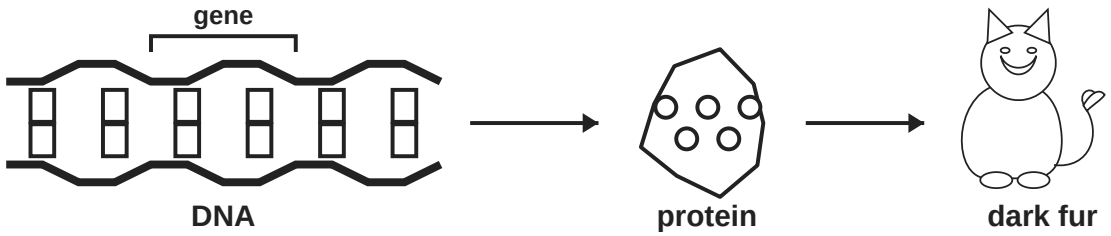
Glue in: Siamese cat data

GLUED IN: SIAMESE CAT DATA

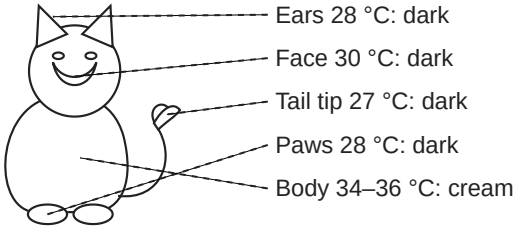
Circled ear (28), face (30), paw (28), tail tip (27) = C; back (36), belly (36), upper leg (34) = W. Litters 1–2: Siamese × Siamese = all Siamese. Litters 3–4: solid parents had Siamese kittens, so genes come from both parents.

The Siamese Cat Mystery (continued)

SketchNotes: DNA to Proteins



Sensemaking: The Siamese Cat Path



Color the points dark.

Newborn kittens are white: warm all over before birth.

Exit Ticket

Dark fur grows on the cool parts of the cat (ears, face, paws, tail; 27–30 °C). The pattern is inherited from both parents, so it starts in the DNA: DNA → gene → protein → dark fur.

Glue in: Reading: DNA, genes, and chromosomes

GLUED IN: READING: DNA, GENES, AND CHROMOSOMES
Underlined DNA, gene, protein, chromosome. Arrow DNA → gene → protein. ? example: How can two solid parents have a Siamese kitten?

What Is DNA Made Of?

Focus question: What is DNA made of?

Vocabulary

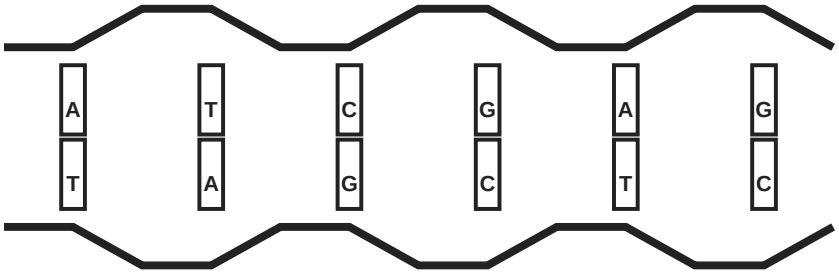
Word	Definition (my words)	This reminds me of… (connections)	Example	Drawing
nucleotide	One building block of DNA: a sugar, a phosphate, and a base.	A link in a chain.	One A nucleotide.	
base	The part of a nucleotide that is a letter: A, T, C, or G. A pairs with T, C with G.	Letters of an alphabet.	A–T, C–G.	 A pairs with T
double helix	DNA's shape: two strands twisted like a ladder.	A twisted ladder or spiral staircase.	A DNA molecule.	

Glue in: Reading: What is DNA made of?

GLUED IN: READING: WHAT IS DNA MADE OF?
Pentagon by sugar, circle by phosphate, rectangle by base. Boxed: A pairs with T. C pairs with G. The cat sequences differ at the 4th letter (G vs. A).

What Is DNA Made Of? (continued)

SketchNotes: DNA Structure



backbone: sugar + phos

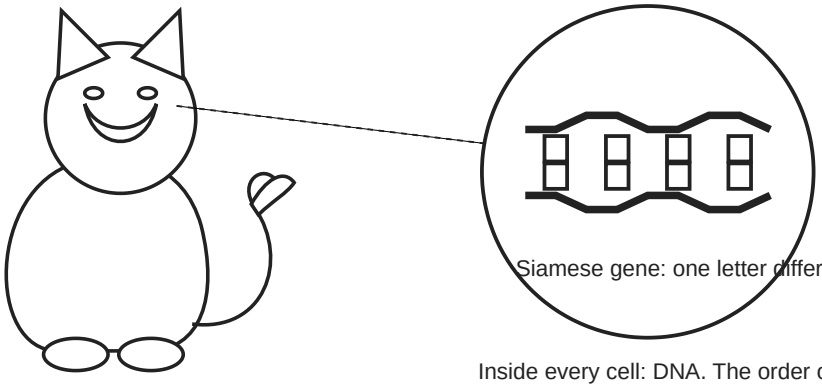
rungs: base pairs

A–T, C–G (4 colors)



1 nucleotide

Sensemaking: The Siamese Cat Path



Siamese gene: one letter different

Inside every cell: DNA. The order of the bases is the message.

Exit Ticket

DNA is a double helix made of nucleotides. Each nucleotide has a sugar, a phosphate, and a base. A pairs with T and C pairs with G. The order of the bases is the message.

What Can One Letter Do?

Focus question: Does every change in DNA matter?

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)	Example	Drawing
base	A DNA letter. Three bases in a row code for one amino acid.	Three letters spell a word.	GGA codes for Gly.	 A pairs with T
amino acid	A small building block linked in a chain to make a protein.	Beads on a necklace.	Gly, Arg, Glu.	
protein	A folded chain of amino acids that does a job.	A machine built from parts.	The pigment-making protein.	

Glue in: Card sort

GLUED IN: CARD SORT

A = 1 (Gly → Arg); B = 3 (done); C = 1 (Leu → Pro); D = 3 (no gene); E = 2 (protein not made); F = 1 (Glu → Val).

What Can One Letter Do? (continued)

SketchNotes: Bases to Amino Acids

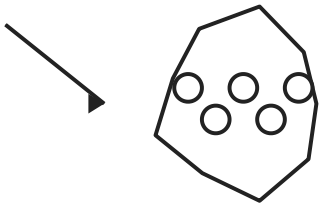
GGA → Gly

AGA → Arg

(one letter changed)

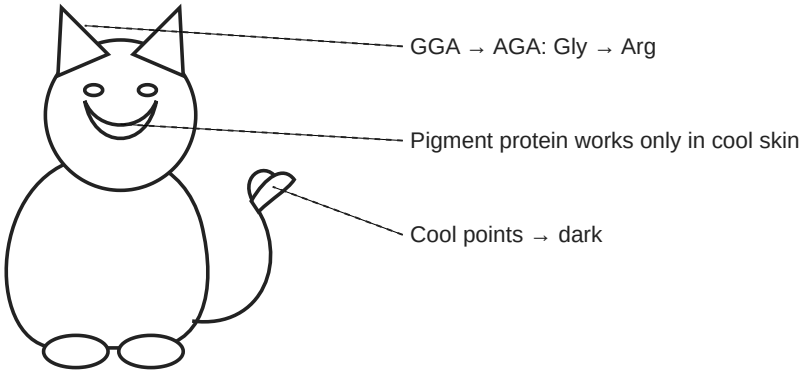


amino acids link in order



protein (folds)

Sensemaking: The Siamese Cat Path



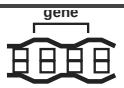
Exit Ticket

Not every change matters. One letter can change an amino acid (Siamese: Gly → Arg), stop a protein from being made, or change nothing (GAG and GAA both make Glu).

Same DNA, Different Jobs

Focus question: Why are only some parts of a Siamese cat dark?

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)	Example	Drawing
gene	A section of DNA for one protein. Every cell has every gene.	Every student has the same textbook.	The pigment gene is in every cat cell.	
protein	What a gene codes for. The proteins a cell makes decide its job.	Tools in a toolbox.	Pigment protein in pigment cells.	
job	The work a cell does, decided by the proteins it makes.	A job at school.	Nerve cells carry signals.	

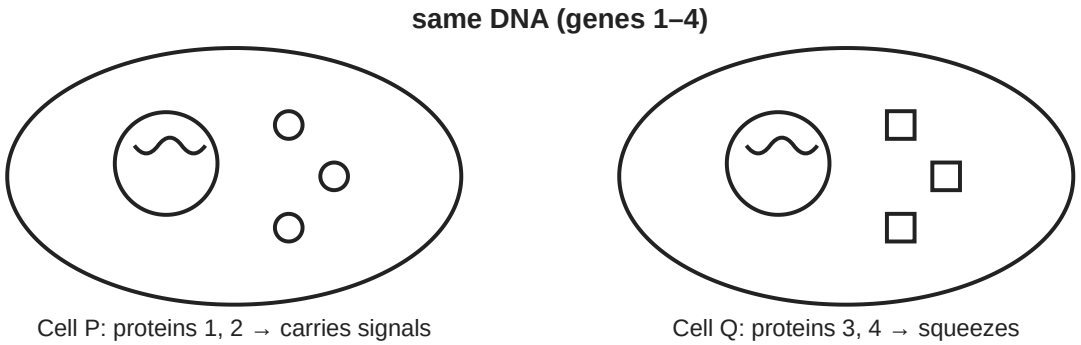
Glue in: Same DNA, different jobs

GLUED IN: SAME DNA, DIFFERENT JOBS

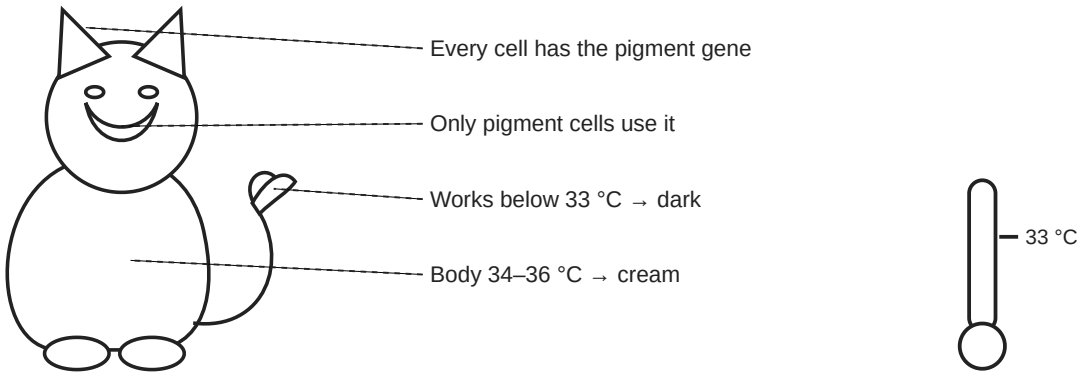
Highlighted: genes 1, 2, 3, 4 in both cells. Table: ear, nose, paw, tail tip = yes, dark; upper leg, back, belly = no, cream.

Same DNA, Different Jobs (continued)

SketchNotes: Same DNA, Different Jobs



Sensemaking: The Siamese Cat Path



Exit Ticket

Only the cool parts are dark. Every cell has the pigment gene, but only pigment cells use it, and the changed protein works only below about 33 °C, so pigment is made only in the ears, nose, paws, and tail.

Case Data

Focus question: How can one DNA letter change the body?

Vocabulary

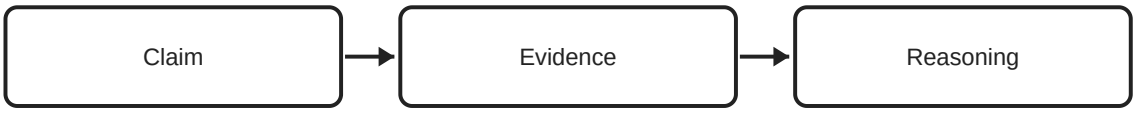
Word	Definition (my words)	This reminds me of… (connections)	Example	Drawing
claim	A one-sentence answer to the question.	Your answer on a test.	A one-letter change caused the dark points.	The answer
evidence	Data, with numbers, that supports the claim.	Proof.	The ear is 28 °C and dark.	28 °C → dark
reasoning	The science idea that links the evidence to the claim.	Explaining why.	One letter → one amino acid → protein works only in cool skin.	because… →

Glue in: Case data

GLUED IN: CASE DATA
Circled: “One letter different” / “One base changed.” Underlined: “One amino acid different.” Boxed: dark points (cat) and “Yes” symptoms (Case 2).

Case Data (continued)

SketchNotes: Claim, Evidence, Reasoning

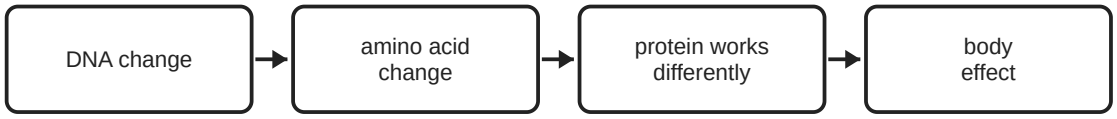


Claim: a one-letter change in the pigment gene.

Evidence: ear 28 °C dark; back 36 °C cream.

Reasoning: one amino acid changes, so the protein works only in cool skin; pigment only on cool parts.

Sensemaking: The Case Path



Cat: one letter → Gly to Arg → works only in cool skin → dark points

Case 2: one base → one amino acid → works poorly → symptoms

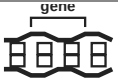
Exit Ticket

One DNA letter changes one amino acid. The protein then works differently or poorly, and that changes the body (dark points in the cat, symptoms in Case 2).

Review and Back to the Cat

Focus question: Why are a Siamese cat's points dark?

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)	Example	Drawing
DNA	Stores the instructions for proteins.	A recipe book.	In every cell.	
gene	A section of DNA for one protein.	One recipe.	Pigment gene.	
protein	A chain of amino acids that does a job.	A worker.	Pigment protein.	
amino acid	A protein building block, set by 3 bases.	A bead.	Gly, Arg.	

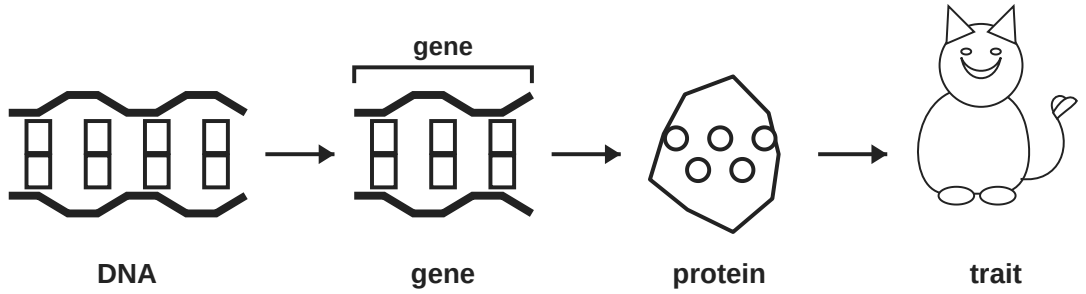
Glue in: Write It: two Siamese kittens

GLUED IN: WRITE IT: TWO SIAMESE KITTENS

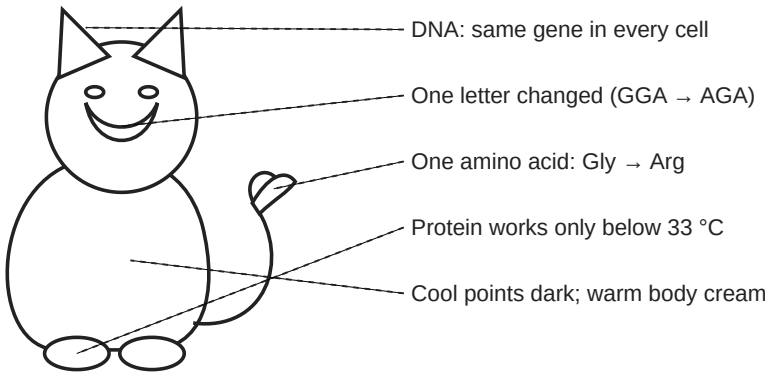
Circled: cool-room ear 26 °C, dark brown; warm-room ear 34 °C, light tan. All 5 model checkboxes checked.

Review and Back to the Cat (continued)

SketchNotes: DNA to Trait



Sensemaking: The Siamese Cat Path



Exit Ticket

A Siamese cat's points are dark because one changed DNA letter swaps one amino acid in the pigment-making protein. That protein works only in skin cooler than about 33 °C, so pigment is made only on the cool ears, face, paws, and tail.